

Quiz 2

Student Name: _____

Student ID: _____

A continuous random variable Y has the probability density function:

$$f(y) = \begin{cases} k(1-y), & 0 \leq y \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

Tasks (5 points each)

1. Find the constant k .

2. Determine the cumulative distribution function $F(y)$.

3. Compute $P(0.25 \leq Y \leq 0.75)$.

4. Find the mean $E[Y]$ and variance of Y .